MTH309 PRACTICE PROBLEMS WEEK 3

Question 1. Find an example of a λ-system which not a σ-field.

Question 2. Find an example of a Monotone class which not a σ-field.

Question 3. (a) If a collection $\mathcal{F}$ of subsets is both a π-system and a λ-system, then show that it is a σ-field.

(b) If a collection $\mathcal{F}$ of subsets is both a field and a Monotone class, then show that it is a σ-field.

Question 4. (Uniqueness of extension)

(a) Let $\mathcal{F}$ be a π-system on Ω. Suppose that $\mathbb{P}_1$ and $\mathbb{P}_2$ are two probability measures defined on the measurable space $(\Omega, \sigma(\mathcal{F}))$ such that they agree on $\mathcal{F}$, i.e.

$$\mathbb{P}_1(A) = \mathbb{P}_2(A), \forall A \in \mathcal{F}.$$

Show that the measures agree on $\sigma(\mathcal{F})$.

(b) Let $\mathcal{F}$ be a field on Ω. Suppose that $\mathbb{P}_1$ and $\mathbb{P}_2$ are two probability measures defined on the measurable space $(\Omega, \sigma(\mathcal{F}))$ such that they agree on $\mathcal{F}$, i.e.

$$\mathbb{P}_1(A) = \mathbb{P}_2(A), \forall A \in \mathcal{F}.$$

Show that the measures agree on $\sigma(\mathcal{F})$, i.e. $\mathbb{P}_1(A) = \mathbb{P}_2(A), \forall A \in \sigma(\mathcal{F})$.

Question 5. Let Ω be a countably infinite set, and let $\mathcal{F}$ be the field consisting of all finite subsets of Ω and their complements. Define

$$\mu(A) := \begin{cases} 0, \text{ if } A \text{ is finite} \\ 1, \text{ if } A^c \text{ is finite} \end{cases}.$$

Show that μ is finitely additive but not countably additive on $\mathcal{F}$, i.e. $\mathbb{P}_1(A) = \mathbb{P}_2(A), \forall A \in \sigma(\mathcal{F})$.

Question 6. (a) Let $f : \mathbb{R} \to \mathbb{R}$ be a function. Show that f is continuous if and only if $f^{-1}(U)$ is open in $\mathbb{R}$ for all open sets U in $\mathbb{R}$.

(b) Let $f : \mathbb{R}^n \to \mathbb{R}^m$ be a function. Show that f is continuous if and only if $f^{-1}(U)$ is open in $\mathbb{R}^n$ for all open sets U in $\mathbb{R}^m$.

Question 7. For any subset A of Ω, define the indicator function of A to be $1_A : \Omega \to \mathbb{R}$ given by

$$1_A(\omega) := \begin{cases} 1, \text{ if } \omega \in A, \\ 0, \text{ otherwise.} \end{cases}$$

Let $\{A_n\}_n$ be a sequence of subsets of Ω. For all $\omega \in \Omega$, prove the following statements.

$$\liminf_{n\to\infty} 1_{A_n}(\omega) = 1_{\liminf_{n\to\infty} A_n}(\omega),$$

and

$$\limsup_{n\to\infty} 1_{A_n}(\omega) = 1_{\limsup_{n\to\infty} A_n}(\omega).$$

Question 8. Let $E \in \mathbb{B}_{\mathbb{R}}$. Say that $A \subseteq E$ is open in E if there exists open $B \subseteq \mathbb{R}$ such that $A = B \cap E$. Show that

$$\mathbb{B}_E := \sigma(\text{all open sets in } E) = \mathbb{B}_{\mathbb{R}} \cap E.$$

Question 9. Let $X : (\Omega, \mathcal{F}, \mathbb{P}) \to (\mathbb{R}, \mathbb{B}_{\mathbb{R}})$ be an RV. Consider the collection of sets

$$\sigma(X) := \{X^{-1}(A) : A \in \mathbb{B}_{\mathbb{R}}\}.$$

Check that it is a σ-field.

Question 10. Let $f : \Omega \to \Omega'$ be a function and let $\mathcal{F}$ be a collection of subsets of Ω'. Show that

$$\sigma(f^{-1}(\mathcal{F})) = f^{-1}(\sigma(\mathcal{F})),$$

where $f^{-1}(A)$ denotes the pre-image of $A \subset \Omega'$, $f^{-1}(\mathcal{F}) := \{f^{-1}(A) : A \in \mathcal{F}\}$ and $f^{-1}(\sigma(\mathcal{F})) := \{f^{-1}(A) : A \in \sigma(\mathcal{F})\}$. (Hint: Use the principle of good sets.)